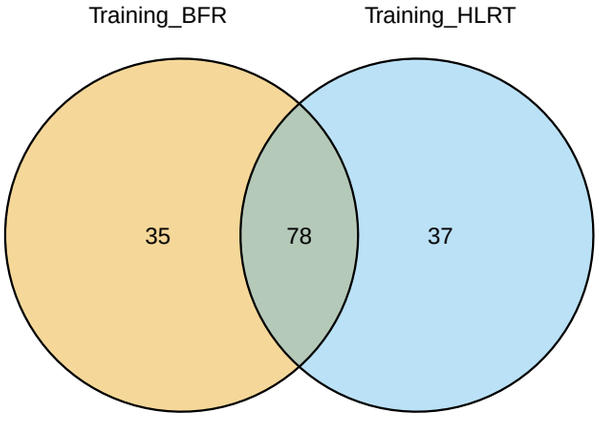
Aerobic respiration and respiratory electron transport

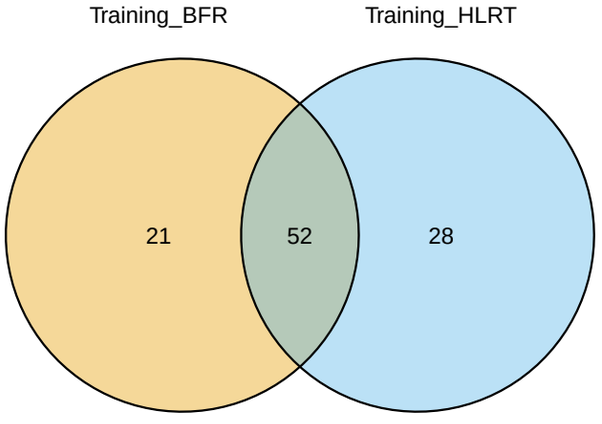
Training_BFR: NES 3.01, padj 1.4e−23
Training_HLRT: NES 2.88, padj 9.6e−23



Shared: ACAD9, ACO2, ATP5F1A, ATP5F1C, ATP5F1D, ATP5MG, ATP5PB, ATP5PD, ATP5PO, C12orf73, COQ10A, COX17, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7B, CS, CYC1, DLAT, DLST, ETFA, ETFB, FAHD1, FH, GOT2, HCCS, HSPA9, IDH3A, IDH3B, IDH3G, LETM1, LYRM7, MDH2, ME2, MT−CO3, NDUFA4, NDUFA9, NDUFAF2, NDUF...

Respiratory electron transport

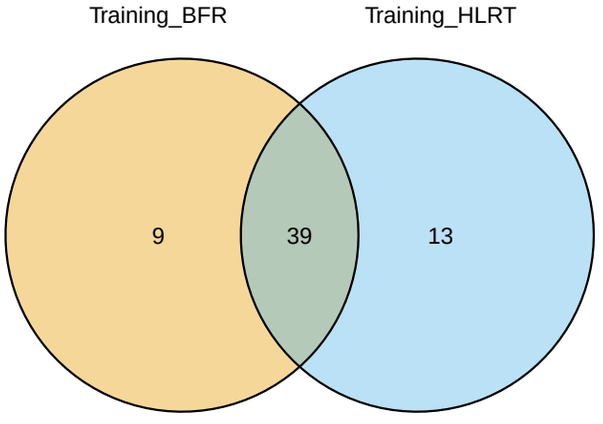
Training_BFR: NES 2.94, padj 1.1e−17
Training_HLRT: NES 2.91, padj 1.2e−18



Shared: ACAD9, C12orf73, COQ10A, COX17, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, COX7B, CYC1, ETFA, ETFB, GOT2, HCCS, HSPA9, ISCU, LETM1, LYRM7, MDH2, MT−CO3, NDUFA4, NDUFA9, NDUFAF2, NDUFAF3, NDUFAF4, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS6, ...

Mitochondrial protein degradation

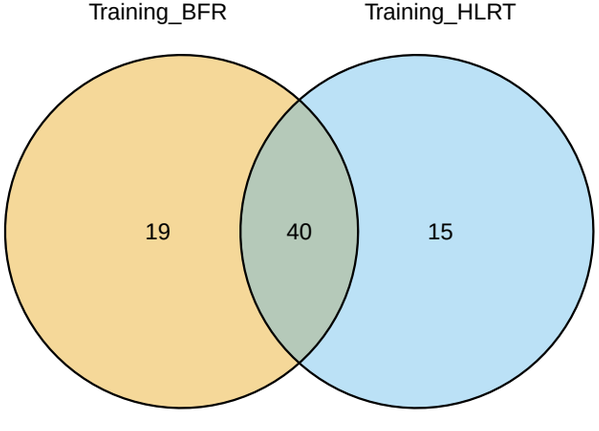
Training_BFR: NES 2.94, padj 3.3e−15
Training_HLRT: NES 2.77, padj 3.4e−13



Shared: ACO2, AFG3L2, ALDH1B1, ATP5F1A, ATP5F1C, ATP5MG, ATP5PD, ATP5PO, CHCHD2, CLPP, COX5B, CS, ECH1, FH, GLUD1, HSD17B10, HSPA9, HSPD1, HTRA2, IARS2, IDH3A, LDHD, LONP1, MDH2, ME2, MRPL12, NADK2, NDUFS1, NDUFS3, NDUFV1, OGDH, OPA1, PCCB, PDHA1, PMPCA, SHMT2, SSBP1, SUCLG2, UQCRC2

Mitochondrial translation

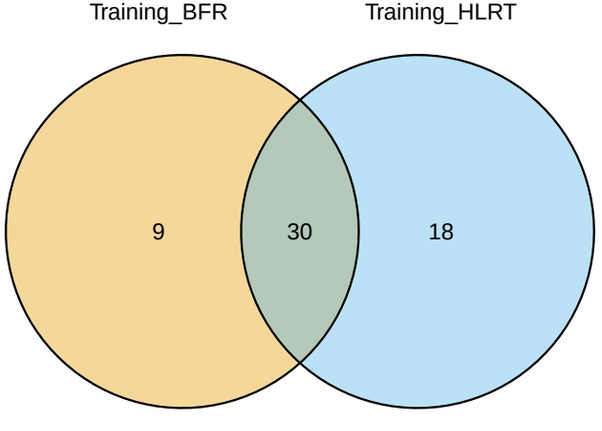
Training_BFR: NES 2.48, padj 3.1e−08
Training_HLRT: NES 2.61, padj 1.4e−10



Shared: DAP3, GFM1, MALSU1, MRPL1, MRPL11, MRPL12, MRPL13, MRPL14, MRPL15, MRPL2, MRPL22, MRPL33, MRPL38, MRPL39, MRPL40, MRPL44, MRPL46, MRPL49, MRPL50, MRPL53, MRPL9, MRPS16, MRPS18B, MRPS22, MRPS23, MRPS24, MRPS27, MRPS28, MRPS30, MRPS34, MRPS35, MRPS9, MRRF, MT−CO3, MT−ND1, NDUFAB1, OXA1L, PTCD3, TSF...

Mitochondrial translation elongation

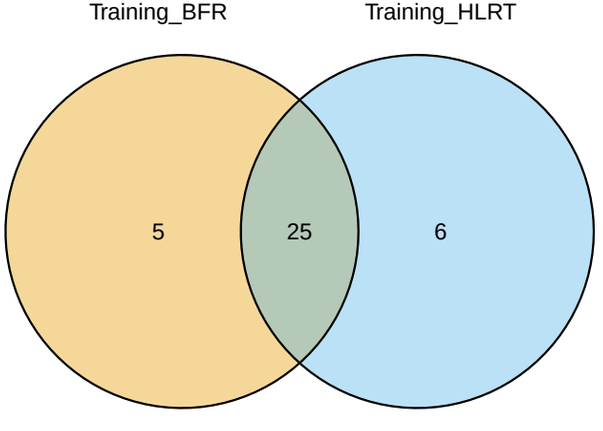
Training_BFR: NES 2.52, padj 4e−08
Training_HLRT: NES 2.63, padj 2.9e−10



Shared: DAP3, GFM1, MRPL1, MRPL11, MRPL12, MRPL13, MRPL14, MRPL15, MRPL2, MRPL22, MRPL33, MRPL38, MRPL39, MRPL44, MRPL46, MRPL50, MRPL53, MRPL9, MRPS16, MRPS18B, MRPS23, MRPS24, MRPS27, MRPS28, MRPS30, MRPS9, OXA1L, PTCD3, TSFM, TUFM

Mitochondrial protein import

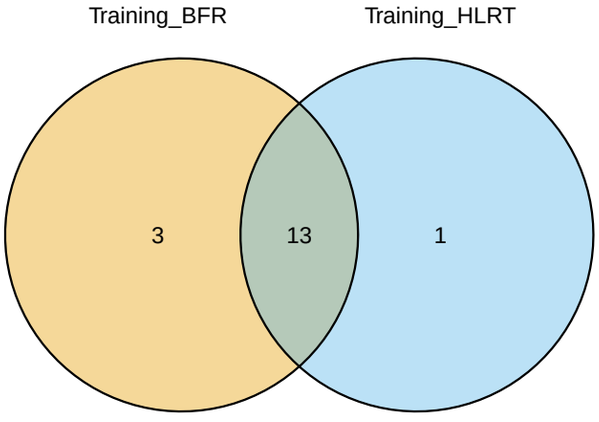
Training_BFR: NES 2.80, padj 3.8e−11
Training_HLRT: NES 2.47, padj 6.2e−08



Shared: ACO2, ATP5F1A, CHCHD2, CMC2, COA4, COA6, COX17, CS, CYC1, GRPEL1, HSPA9, HSPD1, IDH3G, LDHD, MTX1, PITRM1, PMPCA, PMPCB, SAMM50, SLC25A12, SLC25A4, TIMM13, TIMM44, TIMM50, TIMM8A

Citric acid cycle (TCA cycle)

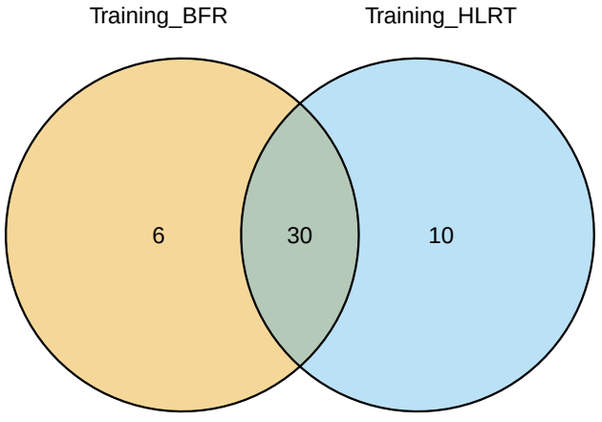
Training_BFR: NES 2.52, padj 1.7e−06
Training_HLRT: NES 2.39, padj 7.9e−06



Shared: ACO2, CS, DLST, FH, IDH3A, IDH3B, IDH3G, MDH2, NNT, OGDH, SUCLA2, SUCLG1, TRAP1

Complex I biogenesis

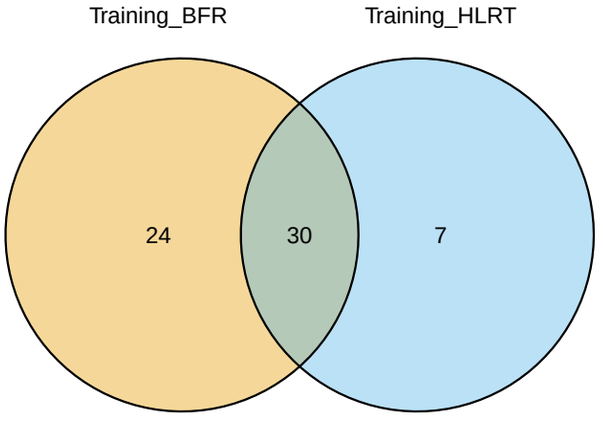
Training_BFR: NES 2.31, padj 1.2e−05
Training_HLRT: NES 2.68, padj 1.4e−10



Shared: ACAD9, HSPA9, MT−ND1, NDUFA10, NDUFA12, NDUFA8, NDUFA9, NDUFAB1, NDUFAF2, NDUFAF3, NDUFAF4, NDUFB1, NDUFB10, NDUFB11, NDUFB4, NDUFB5, NDUFB6, NDUFB7, NDUFB8, NDUFB9, NDUFC2, NDUFS1, NDUFS3, NDUFS4, NDUFS5, NDUFS6, NDUFS8, NDUFV1, NDUFV2, OXA1L

Neddylation

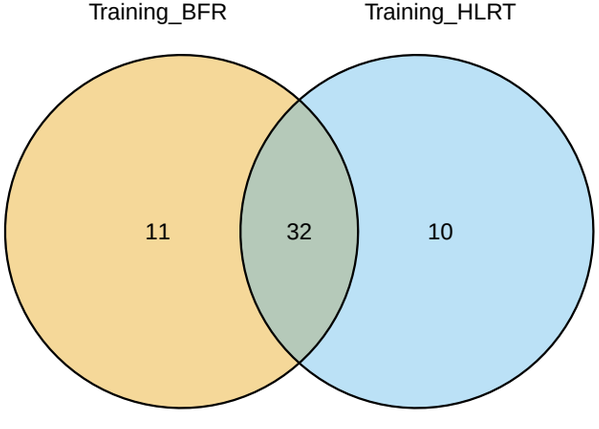
Training_BFR: NES −2.17, padj 1.4e−05
Training_HLRT: NES −2.32, padj 4.3e−07



Shared: ASB11, COPS2, COPS4, COPS7A, COPS8, CUL1, CUL5, FBXO40, GPS1, NAE1, NPLOC4, PSMA1, PSMA2, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMC6, PSMD12, RBBP7, SQSTM1, UCHL3

Protein localization

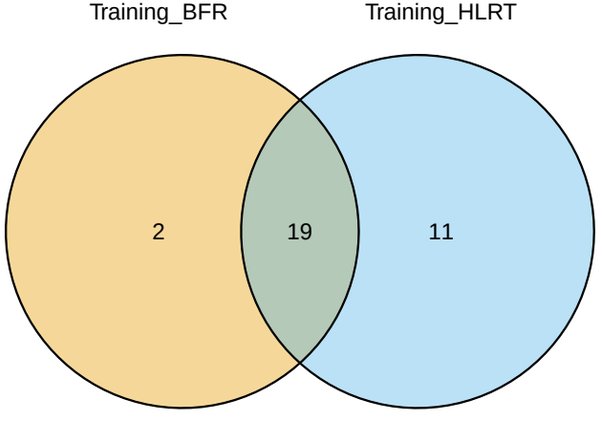
Training_BFR: NES 2.48, padj 4.1e−09
Training_HLRT: NES 2.07, padj 1.6e−05



Shared: ACO2, ATP5F1A, CHCHD2, CMC2, COA4, COA6, COX17, CRAT, CS, CYC1, ECH1, ECI2, EPHX2, GRPEL1, GSTK1, HMGCL, HSPA9, HSPD1, IDH3G, LDHD, MTX1, NUDT19, PITRM1, PMPCA, PMPCB, SAMM50, SLC25A12, SLC25A4, TIMM13, TIMM44, TIMM50, TIMM8A

Signaling by the B Cell Receptor (BCR)

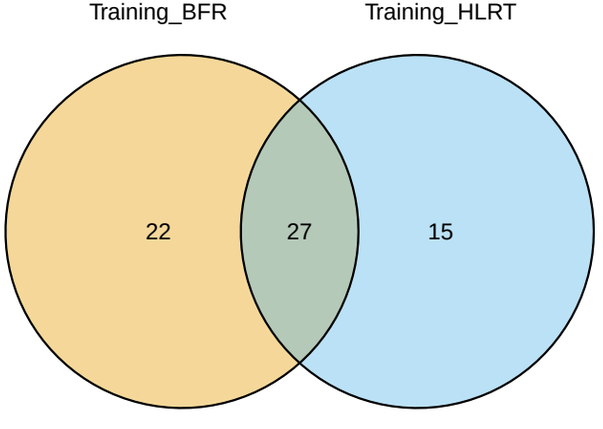
Training_BFR: NES −2.28, padj 2e−05
Training_HLRT: NES −2.43, padj 7.5e−07



Shared: CUL1, GRB2, PIK3R1, PPP3CB, PPP3R1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, STIM1

Antigen processing: Ubiquitination & Proteasome degradation

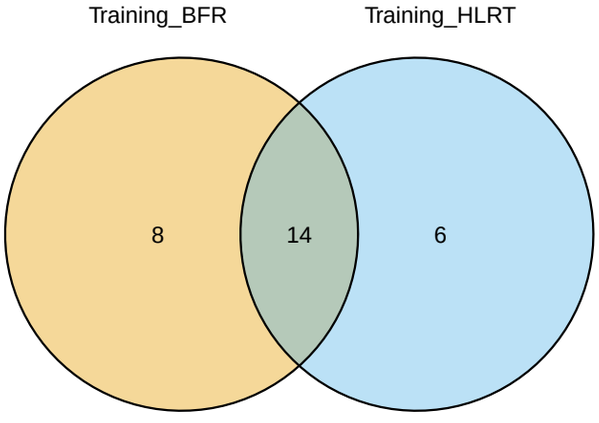
Training_BFR: NES −2.04, padj 7e−05
Training_HLRT: NES −2.44, padj 4.8e−09



Shared: ARIH2, ASB11, BLMH, CUL1, CUL5, HECTD1, HUWE1, NPEPPS, PRKN, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, STUB1, UBE2B, UBE2V2, UBE3A, UFL1

ABC transporter disorders

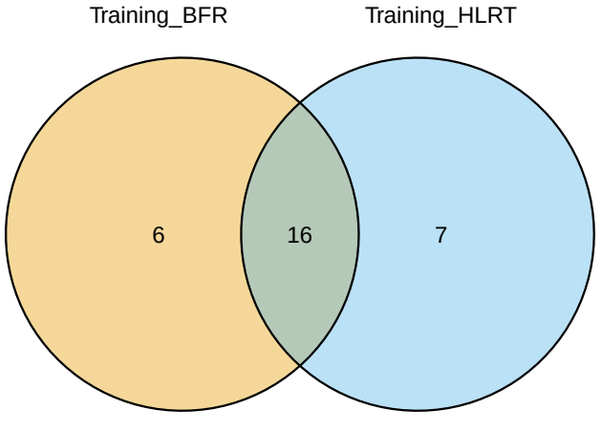
Training_BFR: NES −2.19, padj 8.1e−05
Training_HLRT: NES −2.20, padj 4.1e−05



Shared: ERLIN2, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

G1/S DNA Damage Checkpoints

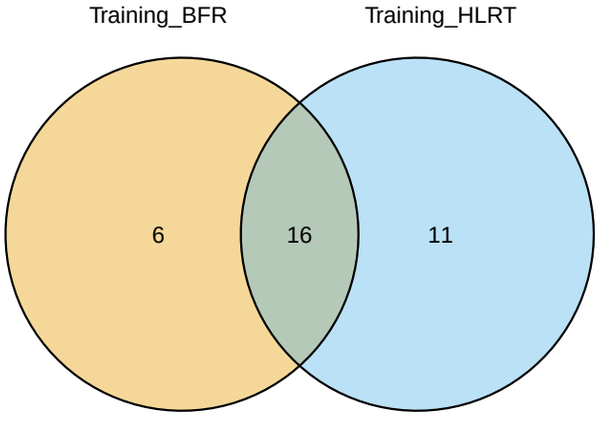
Training_BFR: NES −2.17, padj 9.2e−05
Training_HLRT: NES −2.36, padj 4.5e−06



Shared: CDKN1B, CUL1, MAPK14, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Downstream signaling events of B Cell Receptor (BCR)

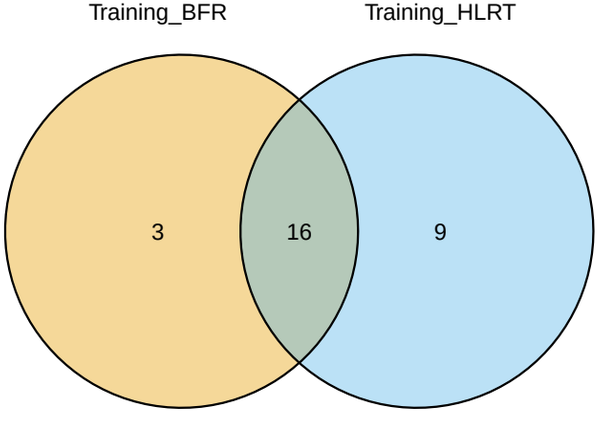
Training_BFR: NES −2.15, padj 0.00011
Training_HLRT: NES −2.35, padj 1.7e−06



Shared: CUL1, PPP3CB, PPP3R1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Synthesis of DNA

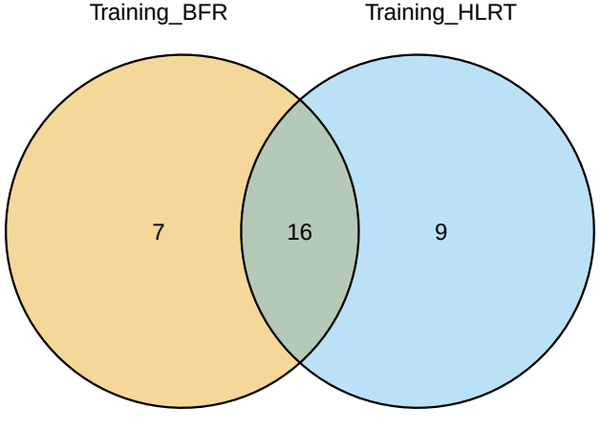
Training_BFR: NES −2.24, padj 0.00014
Training_HLRT: NES −2.22, padj 2.9e−05



Shared: CUL1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD1, RPA1

CLEC7A (Dectin−1) signaling

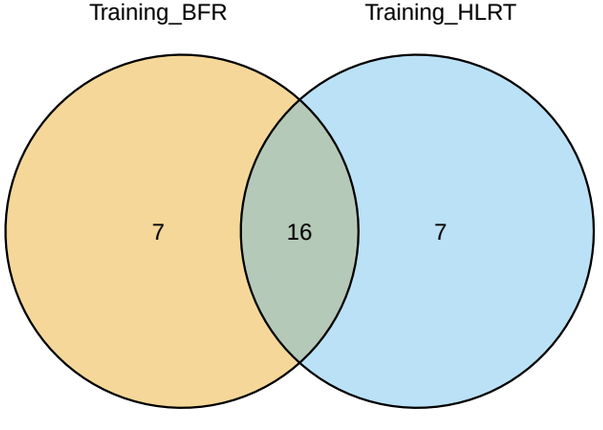
Training_BFR: NES −2.13, padj 0.00016
Training_HLRT: NES −2.42, padj 7.5e−07



Shared: CUL1, PPP3CB, PPP3R1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Degradation of beta−catenin by the destruction complex

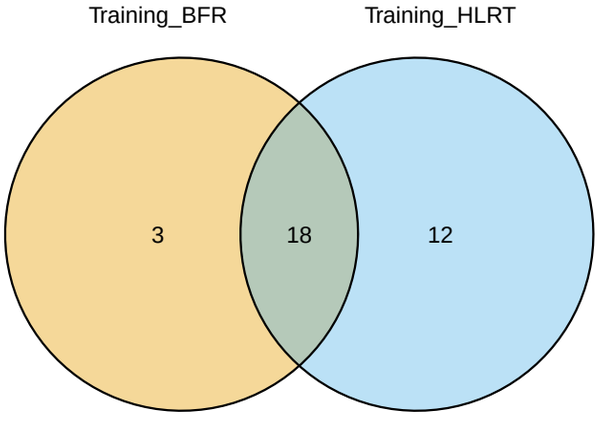
Training_BFR: NES −2.15, padj 0.00016
Training_HLRT: NES −2.14, padj 6.4e−05



Shared: CUL1, PPP2CA, PPP2R5A, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Fc epsilon receptor (FCERI) signaling

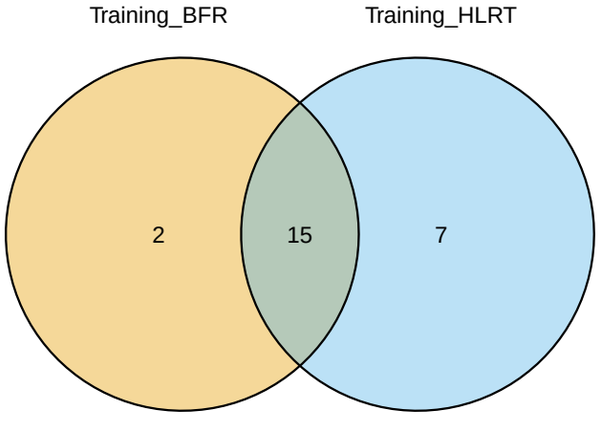
Training_BFR: NES −2.08, padj 0.00023
Training_HLRT: NES −2.27, padj 4.5e−06



Shared: CUL1, GRB2, PIK3R1, PPP3CB, PPP3R1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

SCF(Skp2)−mediated degradation of p27/p21

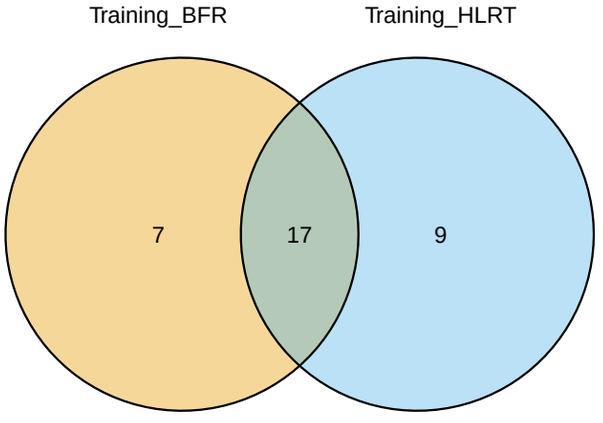
Training_BFR: NES −2.18, padj 0.00023
Training_HLRT: NES −2.39, padj 6e−06



Shared: CDKN1B, CUL1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

S Phase

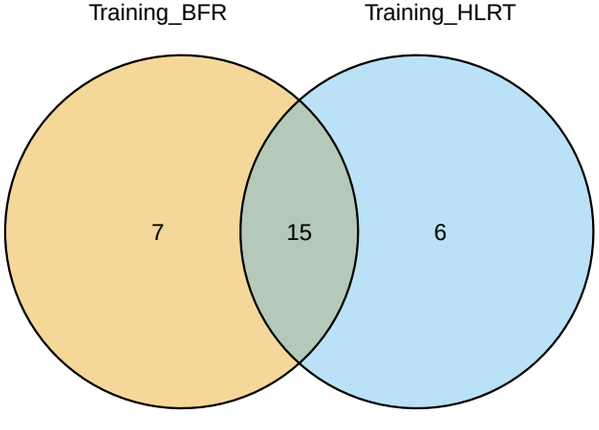
Training_BFR: NES −2.09, padj 0.00023
Training_HLRT: NES −2.03, padj 0.00022



Shared: CDKN1B, CUL1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD1, RPA1

AUF1 (hnRNP D0) binds and destabilizes mRNA

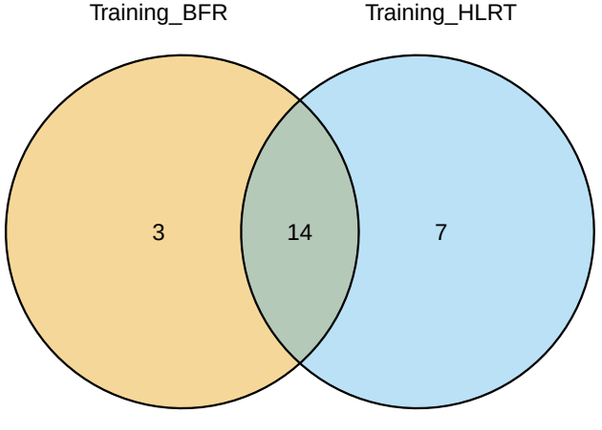
Training_BFR: NES −2.16, padj 0.00025
Training_HLRT: NES −2.31, padj 1.2e−05



Shared: EIF4G1, HSPB1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Degradation of CRY and PER proteins

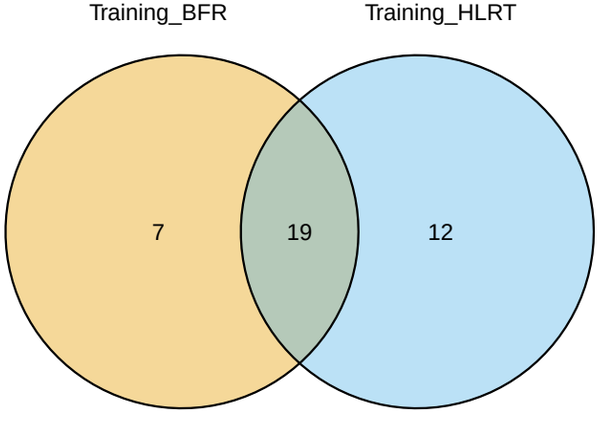
Training_BFR: NES −2.16, padj 0.00025
Training_HLRT: NES −2.30, padj 2.4e−05



Shared: CUL1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Nuclear events mediated by NFE2L2

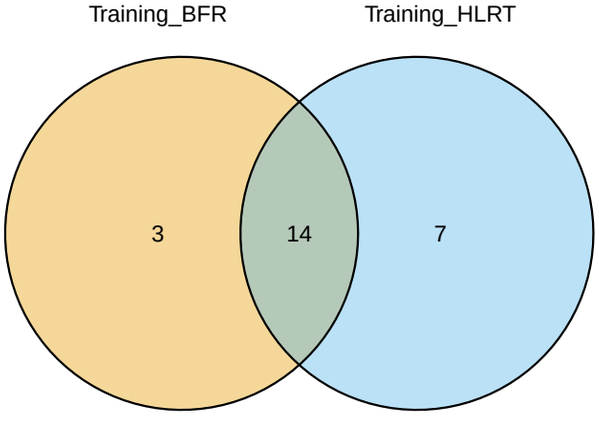
Training_BFR: NES −2.05, padj 0.00025
Training_HLRT: NES −2.21, padj 1.3e−05



Shared: BCL2, CUL1, ME1, PRDX1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD1, SQSTM1

Switching of origins to a post−replicative state

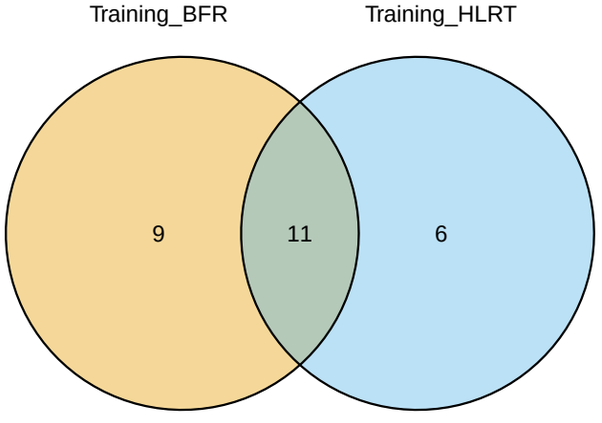
Training_BFR: NES −2.16, padj 0.00025
Training_HLRT: NES −2.30, padj 2.4e−05



Shared: CUL1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Complex IV assembly

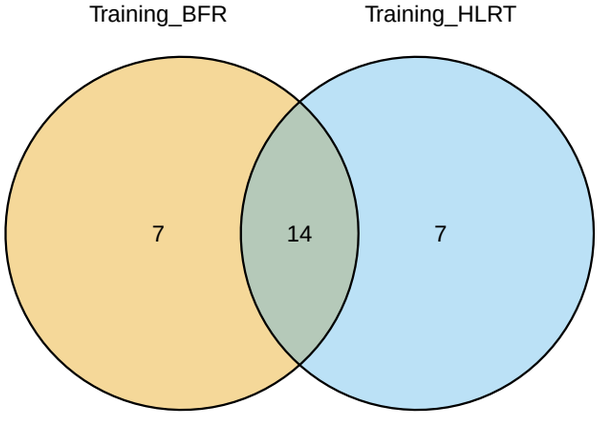
Training_BFR: NES 2.26, padj 0.00022
Training_HLRT: NES 2.15, padj 0.00027



Shared: COQ10A, COX17, COX4I1, COX5B, COX6B1, COX6C, COX7A1, COX7A2, NDUFA4, SMIM20, TACO1

GSK3B−mediated proteasomal degradation of PD−L1(CD274)

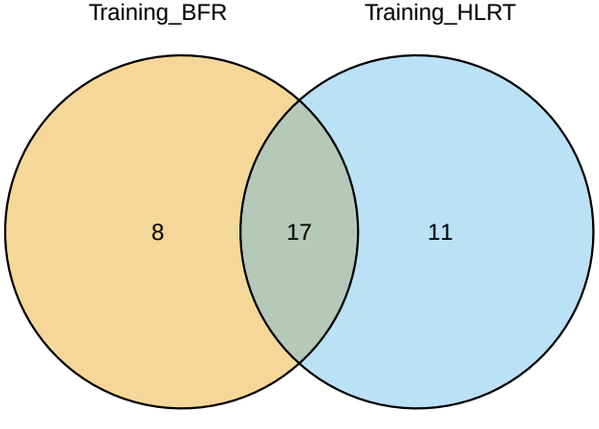
Training_BFR: NES −2.15, padj 0.0003
Training_HLRT: NES −2.26, padj 2.3e−05



Shared: CUL1, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5

Interleukin−1 family signaling

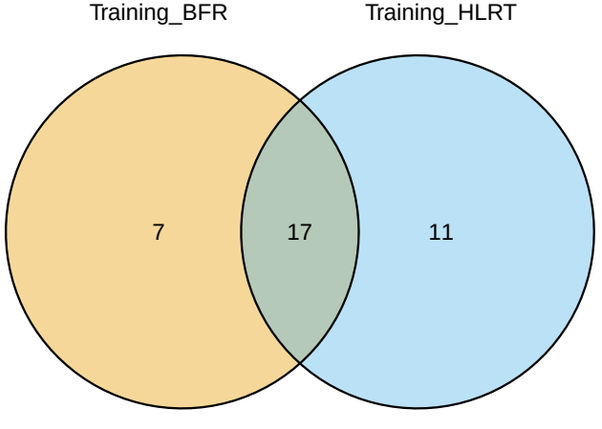
Training_BFR: NES −2.02, padj 0.0003
Training_HLRT: NES −2.38, padj 7.5e−07



Shared: CUL1, MAP2K1, MAP2K6, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SQSTM1

Interleukin−1 signaling

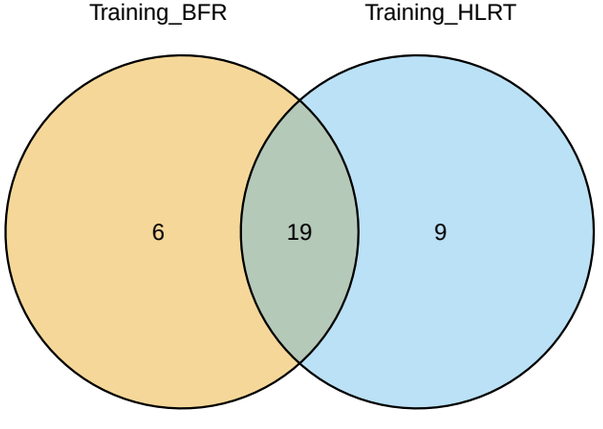
Training_BFR: NES −2.08, padj 0.00031
Training_HLRT: NES −2.41, padj 1.2e−06



Shared: CUL1, MAP2K1, MAP2K6, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, SQSTM1

Mitotic G1 phase and G1/S transition

Training_BFR: NES −2.06, padj 0.00031
Training_HLRT: NES −2.12, padj 6.4e−05



Shared: CDKN1B, CUL1, PPP2CA, PPP2R2A, PSMA1, PSMA3, PSMA4, PSMA5, PSMA6, PSMB1, PSMB2, PSMB3, PSMB4, PSMB5, PSMB6, PSMC2, PSMC5, PSMD1, RPA1